

Abasyn University

Department of Computer Science

Assignment #1

DSA

Subject: DSA

Assignment Number: 1

Submitted To: Amjis

Submitted By: Imran Khan

Registration Number: 21532

Department of: Computer Science

November 20, 2025

Question 1

What is a LinkedList

When we look at data structures, a LinkedList is a pretty fundamental one. Basically, instead of storing elements all in one contiguous block of memory like an array, a LinkedList connects items in a chain. Each item, which we call a "node," holds two main things: the actual data you want to store, and a pointer, or reference, to the next node in the sequence. This pointer is what links everything together.

This way of organizing data is super flexible. You can easily add new nodes anywhere in the list, or remove existing ones, without having to shift a bunch of other elements around. Think of it like a treasure hunt where each clue tells you where the next clue is hidden. You don't need a whole map laid out beforehand; you just follow the links.

What's really cool about this is how it handles memory. Since nodes don't have to be next to each other in memory, you can grow the list dynamically. If you need more space, you just create a new node and link it in. This is a big difference from arrays, which have a fixed size once you create them.

There are a couple of common types, too. You've got singly linked lists, where each node just points to the next one. Then there are doubly linked lists, where each node points to both the next *and* the previous node. This second type makes it easier to traverse the list backwards, which can be handy sometimes. From what I understand, it's all about managing these connections efficiently.